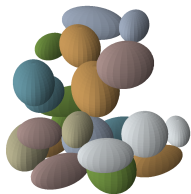
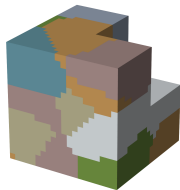


Repair target: `generate_grains()`



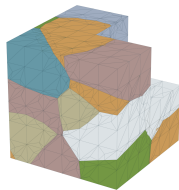
Packed ellipsoids

Kanapy:
`ms.pack()`
Kanapy plot:
`plot_ellipsoids_3D()`



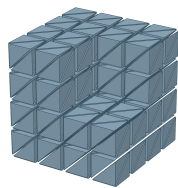
Voxel mesh

Kanapy:
`ms.voxelize()`
Kanapy plot:
`plot_voxels_3D()`



GB surface

Kanapy:
`ms.generate_grains()`
Report plot:
`contextual_rve()`



Bulk mesh (next)

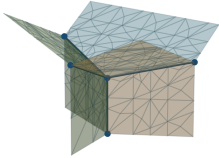
Illustration only.
Report: `mesh_demo()`
Tetrahedra separated
for visibility.

First three: the same RVE; voxels and surfaces share an APD. Fourth: a separate illustration.

Existing volume route: `voxelize()` → `smoothen()` → `write_abq()`

APD → `generate_grains()`: **fix point contacts** → **Bulk mesh (next)**

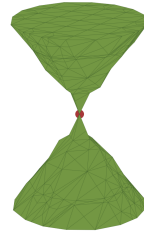
The problem: a grain touches itself



Normal junction

Different grains meet

```
Kanapy geometry: build_grain_geometry()  
Kanapy plot: plot_polygons_3D()
```



Point contact

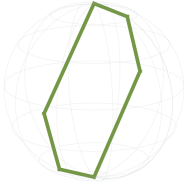
One grain, no finite connection

```
Kanapy geometry: build_grain_geometry()  
Report plot: draw_grain()
```

Goal: watertight grain surfaces from `generate_grains()`.

Controlled examples. The actual failing RVE is still needed.

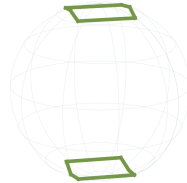
1. Locate the bad connection



Regular vertex

One loop

```
Kanapy surface: build_grain_geometry()  
Report analysis: vertex_link()  
Report plot: link_figure()
```



Point contact

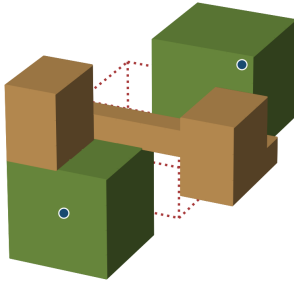
Two separate loops

```
Kanapy surface: build_grain_geometry()  
Report analysis: vertex_link()  
Report plot: link_figure()
```

Planned: check every grain boundary vertex and save defects.

Check volume connectivity separately. Keep normal grain junctions.

2. Change only a small region



Manual voxel example; not an APD repair.

Kanapy plot: `plot_voxels_3D()`

Annotations: report script.

Dashed box

Repair the grain partition.

Try a finite connection.

Adjust neighbours together.

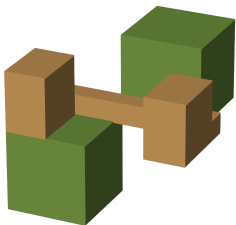
Outside and blue cores

Keep grain labels fixed.

Automatic search: planned

Start with one interior defect. Reject if no candidate passes.

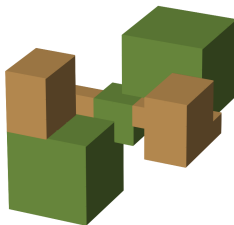
3. Check every affected grain



Before

A touches at a point

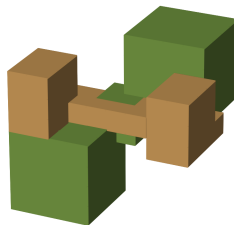
Manual labels. Kanapy plot:
`plot_voxels_3D()`



Change **A** alone

Reject: **B** breaks

Manual labels. Kanapy plot:
`plot_voxels_3D()`



Joint candidate

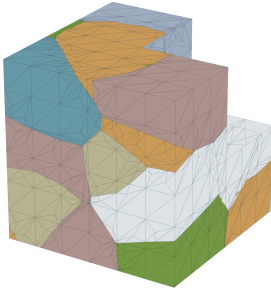
All grains connected

Manual labels. Kanapy plot:
`plot_voxels_3D()`

Also check boundary defects, volume change and shape.

Manual voxel example: 10 of 1331 cells changed. Not a continuous APD repair.

4. Validate the surface for bulk meshing



Current surface output

```
Kanapy: ms.generate_grains()  
Report plot: contextual_rve()
```

First deliverable: a repaired, validated grain boundary surface.

Shown: an illustrative RVE, before any repair or FE validation.

Planned validation

Check closed grain surfaces.

Check surface triangle quality.

Try bulk meshing; check elements.

Mechanics: a later sensitivity check.